Year 13 Further Maths - Applications of matrices

Similar questions - Compositions, powers and areas

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ represent transformations T_A and T_B.

(a) Describe geometrically the single transformation consisting of T_A followed by T_B. **[3]**

(b) By considering T_A, determine $\mathbf{A}^{423}$. **[3]**

$\mathbf{C} = \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$ represents T_C. Region R is $x^2 + y^2 \leqslant 100$ and R' is its image under T_C.

(c) Find the exact area of R' and sketch it, giving the points where its boundary meets the axes. **[5]**

2 $\mathbf{P} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. The triangle T has vertices $(0,0)$, $(2,0)$ and $(0,3)$.

(a) Find the vertices of the image of T under **P** followed by **Q**. **[3]**

(b) Find the single matrix representing **P** followed by **Q** and use its determinant to find the area of the image. **[3]**

(c) Is the area of the image the same if **Q** is applied first? Explain. **[2]**

Question 1

(a) $\mathbf{BA} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: a reflection in the x-axis

(b) T_A is a rotation of 90° anticlockwise about O, so $\mathbf{A}^4 = \mathbf{I}$. $423 = 4 \times 105 + 3$, so $\mathbf{A}^{423} = \mathbf{A}^3 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ (rotation of 270° anticlockwise)

(c) R is a circle of radius 10 and area 100π. $\det \mathbf{C} = \frac{1}{20}$, so the exact area of R' is 5π. R' is the ellipse $\dfrac{x^2}{2.5^2} + \dfrac{y^2}{2^2} \leqslant 1$ (since $x' = \frac{x}{4}$, $y' = \frac{y}{5}$ and $x^2 + y^2 \leqslant 100$ gives $16x'^2 + 25y'^2 \leqslant 100$), meeting the axes at $(\pm 2.5, 0)$ and $(0, \pm 2)$

Question 2

(a) $\mathbf{QP} = \begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$ (apply $\mathbf{P}$ first). $(0,0) \to (0,0)$, $(2,0) \to (0,6)$, $(0,3) \to (-6,0)$

(b) The matrix is $\begin{pmatrix} 0 & -2 \\ 3 & 0 \end{pmatrix}$, determinant 6. Area of T is 3, so the image has area 18

(c) Yes: $\det(\mathbf{PQ}) = \det \mathbf{P} \det \mathbf{Q} = 6 = \det(\mathbf{QP})$, so the area is the same, although the image is in a different position ($\mathbf{PQ} \neq \mathbf{QP}$)